

GemmaScope Position/Budget Checkpoint

Model merging SAE project

Local research log

2026-05-22

Checkpoint reason: position-restricted sparse repair is stable enough to move into feature-ID causality.

Plain Definitions

- Model merging combines model weights or task deltas. In this project, the merge/ablation damaged refusal behavior.
- A sparse autoencoder (SAE) gives sparse feature coordinates that reconstruct Gemma-2-2B MLP activations.
- A patch replaces selected recipient activations with donor-like decoded SAE contributions.
- A position mask asks where the patch must be applied: harmful prompt content, assistant/template boundary tokens, generated tokens, or combinations.

Question: do the selected GemmaScope MLP-SAE features repair refusal because they encode harmful-content semantics, or because they restore a response-start/refusal-state trajectory?

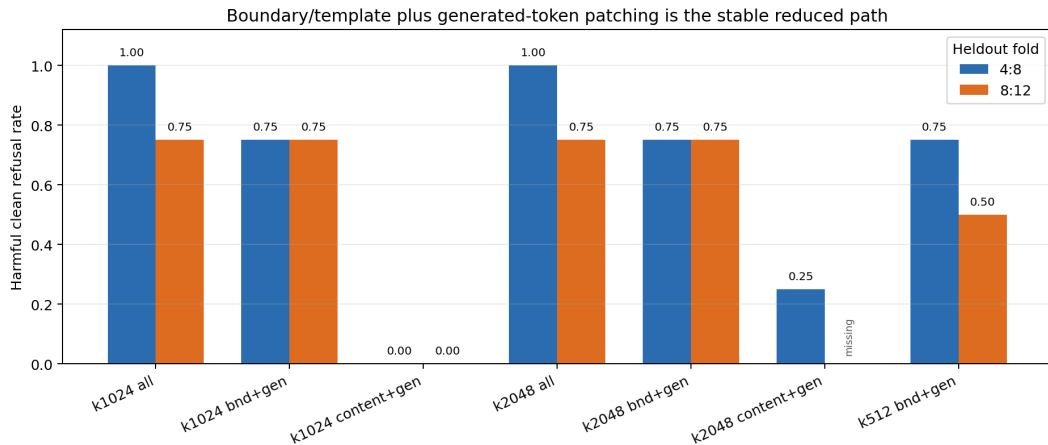
Why it matters:

- A content-feature story would support semantic feature interpretation.
- A trajectory-state story is still mechanistic, but it points to response initialization and rollout maintenance as the merge failure.
- The next feature-level experiment should target the correct causal path.

Key Aggregate Table

Cells are harmful clean refusal / unsafe continuation. Benign helpfulness is 1.00 for all listed rows.

Budget	Fold	All pos	Boundary+gen	Content+gen	Template+gen
k1024	4:8	1.00 / 0.00	0.75 / 0.00	0.00 / 0.00	0.75 / 0.00
k1024	8:12	0.75 / 0.00	0.75 / 0.00	0.00 / 0.00	0.75 / 0.00
k2048	4:8	1.00 / 0.00	0.75 / 0.00	0.25 / 0.25	0.75 / 0.00
k2048	8:12	0.75 / 0.00	0.75 / 0.25	missing	0.75 / 0.25
k512	4:8	–	0.75 / 0.00	–	–
k512	8:12	–	0.50 / 0.00	–	–



Boundary/template plus generated-token patching is the stable reduced path; content+generated stays weak and noisy.

- Static assistant-boundary-only patching fails, so the mechanism is not just one prompt token being overwritten.
- Generated-token-only patching also fails, so rollout maintenance needs a prompt-side seed.
- `assistant_boundary_or_generated` and `prompt_template_or_generated` are the best reduced paths.
- k2048 does not make content+generated competitive; it adds weak repair plus unsafe continuation.
- k512 remains partially causal, so k1024 is not just an overlarge accidental feature set.

Decision And Next Step

Decision: keep the project. This is now a plausible mechanistic claim, but it is not feature-level yet.

Next experiment:

- Move from position masks to feature-ID causality inside the boundary/template plus generated-token path.
- Ablate or isolate top features by layer and generated-token time slice.
- Test whether a compact feature bundle carries refusal-state setup and maintenance.

Provenance

- k1024: `stage3/results/gemma2_2b_gemmaScope_mlp_sae_position_restricted_atomic_v0/`
- k2048: `stage3/results/gemma2_2b_gemmaScope_mlp_sae_position_restricted_k2048_v0/`
- k512 budget: `stage3/results/gemma2_2b_gemmaScope_mlp_sae_boundary_generated_budget_v0/`
- Aggregator: `stage3/scripts/aggregate_gemma2_2b_gemmaScope_mlp_sae_position_restricted.py`
- Copied deck data: `presentations/2026-05-22-1717-gemmaScope-position-budget/data/`